Simulation workflow and ODE solver specifications

The simulation logic is formalized in Algorithm 1. The tripartite replicator dynamics system was solved using MATLAB's built-in `ode45` solver, which implements a variable-step Runge-Kutta (4,5) method. To ensure numerical stability, we enforced a relative tolerance of `RelTol = 1e-6` and an absolute tolerance of `AbsTol = 1e-8` over the integration time span $t \in [0, 100]$. The algorithm details the execution of the primary scenarios and the multi-origin phase space analysis.

Algorithm 1 Simulation workflow for tripartite evolutionary dynamics across multiple scenarios

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Require: Global baseline parameters: $C_{p1}, C_{p2}, u_1, E_m, E_u, C_b, f_{b1}, f_{b2}, R_b, C_u, f_u, R_{p1}, R_{p2}$
Require: ODE Solver settings: Time span $T = [0, 100]$, $RelTol = 10^{-6}$, $AbsTol = 10^{-8}$

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Define scenarios S = {1 : Scenario I, 2 : Scenario II, 3 : Scenario III, 4 : Robustness Check}
Define Replicator Dynamics function F(t, x), where x = [α, β, γ]^T:
    α̇ = α(1 − α) [f_p + γ(1 − β)R_p2 − C_p2]
    β̇ = β(1 − β) [E_m + C_b + γR_b + (γ + αu_2(1 − γ))f_b1 + αu_2 f_b2 − (1 − u_1)(1 − αu_2)E_u]
    γ̇ = γ(1 − γ) [(1 − u_2 α)(1 − β)f_u − C_u]
for each scenario ∈ S do
    Switch scenario do
        Case 1: f_p ← 30, u_2 ← 0.4          ▷ Weak penalty, weak tech (Trap)
        Case 2: f_p ← 400, u_2 ← 0.3         ▷ Strong penalty, weak tech (Limit Cycle)
        Case 3: f_p ← 120, u_2 ← 0.9         ▷ Strong penalty, strong tech (Ideal)
        Case 4: f_p ← 30, u_2 ← 0.4          ▷ Baseline parameters for robustness
    End Switch
    if scenario ≤ 3 then                     ▷ Execute standard 2D time-evolution analysis
        Set initial condition: x_0 ← [0.3, 0.5, 0.4]^T
        X(t) ← ODE45(F, T, x_0, RelTol, AbsTol)
        Plot 2D evolutionary trajectories: α(t), β(t), γ(t) over T
    else                                     ▷ Execute 3D phase diagram for robustness check
        Define initial condition set IC ← {[0.3, 0.5, 0.4]^T, [0.1, 0.3, 0.2]^T, [0.5, 0.5, 0.5]^T, [0.7, 0.7, 0.6]^T}
        Initialize 3D phase plot
        for each x_0 ∈ IC do
            X(t) ← ODE45(F, T, x_0, RelTol, AbsTol)
            Plot 3D trajectory of X(t) in (α, β, γ) space
            Mark starting point x_0
        end for
        Mark evolutionary stable strategy (ESS) convergence point E_2(0, 0, 1)
    end if
end for
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